

Article

Shelf Life Study of Chilled Mullet (*Mugil cephalus*): Histamine Formation and Quality Degradation at Constant and Dynamic Storage Conditions

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Abstract: The present work aimed to evaluate and mathematically model the effect of temperature on *Morganella morganii* growth and histamine formation in farmed mullet (*Mugil cephalus*) during refrigerated storage (at constant temperatures, $T = 0, 2.5, 5, 10$, and $15\text{ }^{\circ}\text{C}$) and to validate the developed models at non-constant temperature conditions (effective temperature $T_{\text{eff}} = 7.4\text{ }^{\circ}\text{C}$). Shelf life evaluation of chilled mullet was also carried out based on microbial spoilage, sensory degradation, and total volatile nitrogen (TVB-N) determination. Spoilage of mullet during refrigerated storage was co-dominated by *Pseudomonas* spp. and Enterobacteriaceae growth. Sensory rejection (score 5 for overall impression) and the end of shelf life coincided with a total microbial load of 8 log cfu/g . The shelf life of chilled mullet was estimated at 15, 11, 7, 3, and 1.5 days at 0, 2.5, 5, 10, and $15\text{ }^{\circ}\text{C}$, respectively. At $T\ 0\text{--}5\text{ }^{\circ}\text{C}$, the time of sensory rejection coincided with TVB-N concentrations of $10.2\text{--}12.3\text{ mg}\cdot 100\text{ g}^{-1}$, and at $10\text{--}15\text{ }^{\circ}\text{C}$, the samples were sensorially rejected before TVB-N development. At storage temperatures $< 5\text{ }^{\circ}\text{C}$, sensory rejection was observed well before histamine levels reached a concentration of 50 mg/kg fish flesh. However, when abusive temperatures prevail, histamine should be considered as a risk factor for the human consumption of mullet.

Keywords: mullet; chilled storage; mathematic modelling; microbial growth; sensory properties; histamine

Key Contribution: Histamine is produced by bacterial actions in fish species (*M. morganii* in mullet) that have a naturally high level of histidine. The FAO and WHO have addressed the public health risks of histamine. Histamine fish poisoning (HFP) has been reported as a result of the consumption of inadequately preserved and improperly stored fish and seafood products (e.g., in high or fluctuating temperatures). However, kinetic data on bacterial growth that contributes to histamine formation in fresh fish, such as mullet, remain scarce. The correlation of histamine production (safety) and fish quality degradation (quality) in chilled mullet in the temperature range of the fish cold chain, and shelf life determination (by mathematical modeling), is important for safety and quality assurance.



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1. Introduction

Chilled fish and seafood are highly perishable foods. Fish deterioration mechanisms are driven by a combination of microbial, enzymatic, and biochemical processes that lead to spoilage and the accumulation of toxic metabolites, particularly biogenic amines, thus impacting the shelf life and quality of the products [1]. Total volatile basic nitrogen (TVB-N) is produced through microbial activity as proteins and non-protein nitrogenous compounds are broken down [2]. Among the biogenic amines, histamine is of significant concern, especially in certain species such as tuna, mackerel, and sardines. The production

of histamine is influenced by various factors, including the quality of the raw materials, the methods used during processing, the conditions of handling and storage, and the presence of specific histamine-producing microorganisms [3]. Managing histamine levels in fish products is essential, as elevated levels have been frequently associated with food poisoning incidents [4].

Histamine is a commonly found biogenic amine in scombroid fish and their products [5,6]. It is more likely to be formed in raw, unfrozen fish, and once produced, it remains stable even under heating and freezing conditions [7]. It does not affect the sensory characteristics of fish products (such as appearance, odor, or taste). The primary producers of histamine in fish products belong to the Enterobacteriaceae family, with species such as *Morganella morganii*, *Morganella psychrotolerans*, and *Klebsiella* spp., as well as *Photobacterium* spp., which belong to the Vibrionaceae family. These bacteria produce the enzyme histidine decarboxylase, which converts the naturally occurring histidine in fish muscle into histamine [8–10]. In mullet, studies have identified the presence of histamine-producing bacteria, such as *Photobacterium damsela* subsp. *damsela* in fish captured in the Ligurian Sea [11]. Additionally, the histamine-producing potential of bacteria in fish has been evaluated using biosensors, highlighting the relevance of these bacterial species in histamine formation [12]. Among them, *M. morganii* is particularly prolific, consistently producing high levels of histamine under favorable conditions [13]. Given its optimal growth temperature of around 25 °C, *M. morganii* poses a significant risk in seafood when exposed to abusive temperatures during handling and storage [14].

M. morganii is one of the most important bacteria that can produce histamine in seafood, especially at temperatures above 10–15 °C. It has a higher optimal temperature for growth and histamine formation than other bacteria, such as *Morganella psychrotolerans* and *Photobacterium* spp. The optimal temperature for histamine formation by *M. morganii* is 25 °C, while for *M. psychrotolerans* and *Photobacterium* spp., it is 15 °C and 20 °C, respectively [10]. This means that *M. morganii* can grow faster and produce more histamine than other bacteria at higher temperatures, which are common in seafood processing and storage [14].

In 2012, the FAO and WHO addressed the public health risks of histamine (and other biogenic amines) from fish and fish products. Histamine fish poisoning (HFP) has been reported as a result of the consumption of inadequately preserved and improperly stored fish and seafood products. In healthy individuals, HFP can occur when a dose of at least 50 mg of histamine is consumed in fish and fish products, with symptoms similar to allergic reactions. Ingestion of food containing small amounts of histamine can result in histamine intolerance, whereas histamine intoxication can appear in everyone as a result of its high concentrations in foods such as fish and fishery products. Assuming a single serving size of 250 g of fish as a standard portion for high fish consumption by an adult, a limit of 200 mg histamine per kg of fish has been estimated (which corresponds to the consumption of 50 mg histamine) [15,16]. According to US regulations, seafood is safe for consumption when the histamine concentration is below 50 mg/kg. The value considered in the EU is 100 mg/kg [16]. Commission Regulation (EC) No 2073/2005 sets the appropriate sampling plan and the analytical reference method for histamine in fish products placed on the market at any stage of the cold chain. Currently, the maximum limits of acceptability for histamine concentrations in fish flesh are used to ensure the safety of scombroid fish.

Mullet (*Mugil cephalus*) species are of great ecological and economic importance for both fisheries and aquaculture around the world, which is mainly concentrated in the Mediterranean, the Black Sea region, and Southeast Asia [17]. It is captured during spawning migration in order to harvest the egg roe, which is salted and dried to be sold as a delicacy. In many parts of the world, increased demand for mullet roe raised the species' commercial value [18]. Mullet presents good biological characteristics and positive aquaculture characteristics: the euryhaline capabilities of many mullets, their fast growth, and their high efficiency in converting food to body mass [19]. It is a type of fish that is often consumed raw or fermented in some countries, such as Greece, Italy, and Egypt, and

an important resource for fisheries and aquaculture in many countries, especially in the Mediterranean region. However, it is also susceptible to histamine formation by bacteria such as *M. morganii*, which can cause food poisoning when consumed in high amounts [20]. Mullet stored at high or fluctuating temperatures poses an HFP risk, with free histidine levels between 2060–7600 mg/kg [16]. The histamine levels in these products are generally below the legal limit of 20 mg/kg, but they could increase during fermentation or improper storage conditions [21]. Therefore, it is important to monitor the presence and activity of *M. morganii* in mullet products to ensure food safety and quality. However, there is a lack of kinetic data on the growth of bacteria contributing to the spoilage and shelf life limitations of fresh mullet.

Modeling histamine formation in mullet is particularly useful for developing effective food safety strategies because mullet is often consumed raw or fermented, especially in regions with traditional culinary practices that involve limited cooking or processing. When fish is consumed raw or only lightly processed, the risk of histamine exposure is higher due to the potential for bacterial growth during handling and storage. By modeling the conditions under which histamine formation occurs, such as temperature, microbial load, and time, food safety experts can better predict and control histamine levels in mullet products. Such models can serve as a valuable preventive tool in predicting histamine levels at various stages of the supply chain, allowing for tailored guidelines for handling, storage, and processing. For instance, temperature control strategies can be optimized, and the timing of processing steps can be adjusted to minimize histamine formation. This is especially crucial for regions where mullet is consumed in raw or fermented forms, as the lack of a cooking step means there is no opportunity to reduce or eliminate histamine levels before consumption [22].

De Silvestri et al. [23], who investigated the effects of different storage temperatures on the growth of psychotropic pathogen bacteria in perishable foods, found that temperature is the main factor in controlling the microbial growth in perishable foods, and that the lag phase and growth rate proved to be heavily affected by temperature whereas the inoculum size did not [23]. It has been reported that the temperature conditions of the real cold chain deviate significantly from the recommended range, resulting in significant quality loss at different stages, including retail and domestic storage. The variation in temperature conditions during product distribution and storage can significantly increase the rate of quality degradation of fish and seafood [24]. Temperature conditions in the cold chain determine the risk potential, shelf life, and final quality of chilled fish and seafood. Based on reliable mathematical models for the chemical and sensory quality indices of fish and seafood products, the effect of temperature can be translated to quality status, from production to final consumption.

The objective was the systematic evaluation and mathematical modeling of *M. morganii* growth and histamine formation in mullet during storage at constant ($T = 0, 2.5, 5, 10$, and $15\text{ }^{\circ}\text{C}$) and variable ($T_{\text{eff}} = 7.4\text{ }^{\circ}\text{C}$) temperature conditions. The effect of the storage temperature on microbial spoilage and quality deterioration was additionally investigated in refrigerated conditions.

2. Materials and Methods

2.1. Sample Delivery and Preparation

Mullet (*Mugil cephalus*) was provided by a Greek aquaculture company (V. GEITONAS & Co., Ltd., Arta, Epirus, Greece) and transferred, ice-packed, to the NTUA Laboratory of Food Chemistry and Technology within one day from harvesting under melting ice conditions. Mullet was harvested in tank farms, by first lowering water levels to concentrate the fish and then gently gathering the fish using a seine net. Fish was slaughtered by immersion in iced water (ice shock) and finally packed in melting ice conditions to maintain freshness during transport or processing.

2.2. Inoculation of Minced Mullet Flesh with *M. morganii*

For the mathematical modeling of histamine formation, minced mullet flesh was inoculated with an initial load of *M. morganii* at 2.4 log cfu/g and stored at constant (0, 5, 10, 15, and 20 °C) temperature conditions. An independent experiment was also carried out at variable conditions (5 h at 5 °C, 5 h at 8 °C, and 2 h at 10 °C, $T_{eff} = 7.4$ °C) to validate the developed predictive models under dynamic conditions.

M. morganii ATCC 25830 was used as the inoculum, as it can produce histidine decarboxylase to enzymatically decarboxylate histidine and produce histamine. The freeze-dried culture was activated by two successive transfers in Nutrient Broth (Merck, Darmstadt, Germany) after incubation at 37 °C for 24 h. The microbial load of *M. morganii* was determined in Nutrient Agar, before inoculation of the mullet samples. Autoclaved (121 °C, 15 min) minced fish meat was inoculated with the *M. morganii* inoculum ($No = 2\text{--}2.5$ log cfu/g) to eliminate naturally occurring pathogens and competing endogenous organisms [25–27]. The inoculum was homogeneously dispersed throughout the minced fish flesh by vigorously mixing in a sterile plastic bag using a Stomacher homogenizer for 2 min. The homogeneity of dispersal was evaluated by enumerating *M. morganii* in 5 subsamples of the ground fish meat. Inoculated and non-inoculated (Control) samples were distributed into sterile plastic tubes and stored at 0, 5, 10, 15, and 20 °C, to simulate mild to moderate temperature abuse conditions. At predetermined storage times, the microbial load was measured on Nutrient Agar (Merck, Darmstadt, Germany). Circular, dome-shaped, mucoid, opaque, cream-colored colonies were enumerated (24 h at 37 °C). The Veratox® Histamine test kit (AOAC-RI #070703 approved method, Product No. 9505, Neogen, Lansing, MI, USA) was used for the quantitative analysis of histamine in fish flesh. The test is a competitive direct ELISA. The range of quantitation was from 2.5 mg/kg to 50 mg/kg. The color intensity in the microtiter wells was measured photometrically using the Spectrostar Nano plate reader (BMG labtech, Ortenberg, Germany). The MARS Data Analysis software v.3.01 R2 (BMG labtech, Ortenberg, Germany) was used to evaluate the results.

Mathematical Modeling of *M. morganii* Growth and Histamine Formation

The growth of *M. morganii* and histamine formation were modeled using the Baranyi Growth Model [28–30]. The DMFit program, available at <http://www.combase.cc/index.php/en/> (accessed on 23 September 2024), was applied for curve fitting. The model parameters (i.e., rate k , lag phase λ), were calculated at all tested temperature conditions. The temperature dependence of microbial growth and histamine formation were modeled using the Arrhenius equation (Equation (1))

$$\ln k = \ln k_{ref} - \frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \quad (1)$$

where k , k_{ref} (in d^{-1}) is the rate constant at any temperature T (in K), and at the reference temperature, T_{ref} (4 °C), respectively, R is the universal gas constant, and E_a is the activation energy of the studied action (in J/mol) [31].

The developed mathematical models were validated using an additional experiment at variable conditions ($T_{eff} = 7.4$ °C). The effective temperature T_{eff} integrates, in a single value, the effect of the variable temperature profile at a specific time of the time–temperature scenario. k_{eff} , the value of the rate constant corresponding to the effective temperature (T_{eff}), was estimated by Equation (2), where t_{tot} is the duration of the time–temperature scenario [31].

$$k_{ref} \cdot \sum_i \left[\exp \left(-\frac{E_a}{R} \cdot \left(\frac{1}{T_i} - \frac{1}{T_{ref}} \right) \right) \cdot t_i \right] = k_{eff} \cdot t_{tot} \quad (2)$$

2.3. Shelf Life Evaluation of Mullet

Mullet slices (50 g each) were stored aerobically (not in sealed pouches) at controlled constant conditions (0 °C, 2.5 °C, 5 °C, 10 °C, and 15 °C; ± 0.2 °C) (Sanyo MIR 153, Sanyo Electric, Ora-Gun, Gun-ma, Japan). Quality assessments at predetermined times, depending on the storage temperature, were estimated and kinetically modeled. The temperature was constantly monitored during the storage experiment (COX TRACER®, Belmont, NC, USA). Shelf life determination of fish was based on microbial (total viable count—TVC, Enterobacteriaceae, and *Pseudomonas* spp.), sensory (appearance, odor, and overall impression), and chemical (total volatile basic nitrogen—TVB-N) quality indices.

2.3.1. Microbial Growth Determination

A sample (10 g) was put into a sterile bag (stomacher) with 90 mL of sterilized Ringer solution (Merck, Darmstadt, Germany) and homogenized for 60 s (Stomacher, BagMixer® interscience, France). The surface of the suitable media in Petri dishes was covered with samples (0.1 mL) of 10-fold serial dilutions of fish. After a 72 h incubation period at 25 °C, the total aerobic viable count was measured on Plate Count Agar (PCA, Merck, Darmstadt, Germany). *Pseudomonas* spp. was enumerated on Cetrimide Agar (CFC, Merck, Darmstadt, Germany) after a 48 h incubation at 25 °C. For Enterobacteriaceae enumeration, the pour plate method and Violet Red Bile Glucose Agar (VRBG, Merck, Darmstadt, Germany) were used. The latter was incubated for 24–48 h at 37 °C. Two replicates of at least three appropriate dilutions were enumerated.

2.3.2. Sensory Evaluation

The sensory evaluation of raw fish was conducted by eight trained assessors of the sensory laboratory of the National Technical University of Athens School of Chemical Engineering. The panelists had been trained in sensory analysis according to ISO 8586:2023 standards [32,33]. The assessors obtained a detailed description of the test and were informed about the food samples that would be assessed. First, before participating in the sensory analysis panel, they developed a list of profiling attributes of the raw fish samples of the present study. During sensory testing, assessors were presented with the raw fish slices on 3-digit coded plates (one sample for each table, 3 samples) at room temperature. The sensory parameters, appearance, and odor of the raw fish samples, as well as the overall impression and sensory quality, were evaluated by scoring using appropriate forms. Ratings were assigned separately for each parameter on a 1 (poor quality)–9 (high quality) scale. Five (5) was taken as the sensory score for minimally acceptable quality [24].

Total Volatile Basic Nitrogen (TVB-N) Determination

The Kjeldahl method was used to determine the total volatile basic nitrogen (TVB-N). TCA extraction was carried out once, followed by titration with sulfuric acid and distillation in a Kjeldahl quick distillation unit (Buchi 321 Distillation unit, Flawwil, Switzerland) [34].

2.4. Statistical Analysis

The growth of *M. morganii* and the histamine formation in farmed mullet during refrigerated storage were evaluated by analysis of variance (ANOVA) at a 95% significance level (STATISTICA® 7.0; StatSoft Inc., Tulsa, OK, USA). Duncan's multiple range test was used to determine significant differences. All measurements were performed in duplicate. A *p*-value is a measure used in statistical hypothesis testing to determine the significance of the results. It represents the probability of obtaining results at least as extreme as the observed results, assuming that the null hypothesis is true. A low *p*-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so the null hypothesis is rejected. This suggests that the results are statistically significant, while a high *p*-value (>0.05), indicates weak evidence against the null hypothesis, so the null hypothesis is not rejected.

The comparison between the experimental (actual) and predicted (calculated by the mathematical models) values was based on the accuracy (A_f) and bias (B_f) factors (Equations (3) and (4))

$$A_f = 10^{\frac{\sum |\log(\frac{y_{predicted}}{y_{experimental}})|}{n}} \quad (3)$$

$$B_f = 10^{\frac{\sum \log(\frac{y_{predicted}}{y_{experimental}})}{n}} \quad (4)$$

where n is the number of observations, and the relative error (RE) is calculated by Equation (5) for each one of the obtained y_i values. Perfect agreement between the predicted values and the corresponding observed values is represented by A_f and B_f values of 1 [35].

$$RE = \frac{(y_{observed} - y_{predicted})}{y_{predicted}} \quad (5)$$

3. Results and Discussion

3.1. *M. morganii* Growth and Histamine Formation in Mullet

The results of the growth of *M. morganii* in mullet flesh at constant conditions are shown in Figure 1. Data derived from the experiments conducted were analyzed using the DMFit program to fit the Baranyi growth model. The initial inoculation level of *M. morganii* in the mullet flesh was 2.4 ± 0.1 log cfu/g. No growth was observed at 0 °C within 1 month of storage ($p > 0.05$). Slow growth was observed at 5 °C, with populations after 1 month of storage reaching values of 5.0 ± 0.1 log cfu/g ($p < 0.05$). Histamine production is generally favored at temperatures above 5 °C, with an optimal range of 20–30 °C [9,36]. A significant increase in *M. morganii* counts with storage time was observed at 10, 15, and 20 °C. The maximum *M. morganii* load at 10 °C was 8.9 ± 0.1 log cfu/g ($p < 0.05$). In the case of 15 and 20 °C, the maximum *M. morganii* populations were 9.3–9.5 log cfu/g and were recorded after 6 and 3 days, respectively ($p < 0.05$). Lee et al. [37] observed similar temperature effects on histamine formation, noting that histamine production in milkfish dumpling samples was significantly more rapid at storage temperatures of 25 and 37 °C than at 15 and 4 °C [37]. Increasing temperatures stimulate both bacterial growth and the production of histidine decarboxylases, with the enzyme's activity being favored at temperatures over 15 °C [9,38], which is in accordance with the results of the present work. In this study, no lag phase ($\lambda = 0$) in *M. morganii* growth was observed under all the tested storage temperatures. Kim et al. [36] also reported that the growth of *M. morganii* inoculated into a tuna fish infusion broth led to the production of the highest levels of histamine at 25 °C. These levels decreased with decreases in the temperature, with the lowest histamine levels formed at 15 °C. Neither growth of the bacteria nor formation of histamine was noted at 4 °C, which is in agreement with the results of the present study (no growth at 0 °C, slow growth at 5 °C).

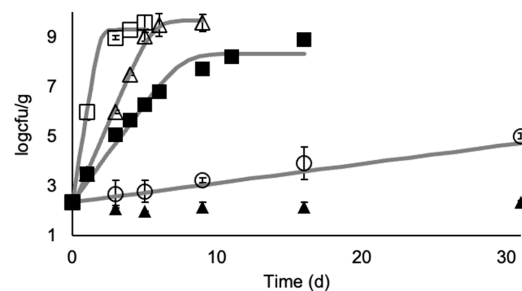


Figure 1. *M. morganii* growth in aerobically packed mullet flesh during isothermal storage at ▲ 0 °C, ○ 5 °C, ■ 10 °C, △ 15 °C, and □ 20 °C (experimental points and predictions are based on the Baranyi Growth Model).

Growth rates of *M. morganii* in mullet flesh increased significantly with storage temperature, as indicated in Table 1 ($p < 0.05$). Kinetic data of *M. morganii* growth were fitted to the Arrhenius equation (Equation (1)) and the E_a and k_{ref} ($T_{ref} = 8\text{ }^{\circ}\text{C}$) values were calculated as 154.3 kJ/mol and 0.285 d^{-1} , respectively.

Table 1. Growth rate (k in d^{-1}) of *M. morganii* in aerobically packed mullet flesh during isothermal storage at 0–20 $^{\circ}\text{C}$. (Average values $\pm$ Standard error).

Temperature ($^{\circ}\text{C}$)	k (d^{-1})
0	–
5	0.098 ± 0.008
10	0.792 ± 0.071
15	1.355 ± 0.064
20	3.615 ± 0.442

In order to validate the developed predictive model for *M. morganii* growth in mullet flesh under variable conditions, an experiment under dynamic conditions (based on the time–temperature scenario presented in Figure 2) was carried out. Figure 2 illustrates the comparison between observed and predicted growth under dynamic conditions. The calculated RE values revealed that all points fall within the acceptable prediction zone of $\pm 20\%$ as defined in the literature [39,40], indicating that the developed mathematical models can be adequately used for the prediction of *M. morganii* growth in chilled mullet at variable temperature conditions. The A_f and B_f values indicated that there was a satisfactory agreement between predicted and observed histamine levels. The B_f values were within the boundaries of 0.7 (fail—safe) to 1.15 (fail—dangerous) [41].

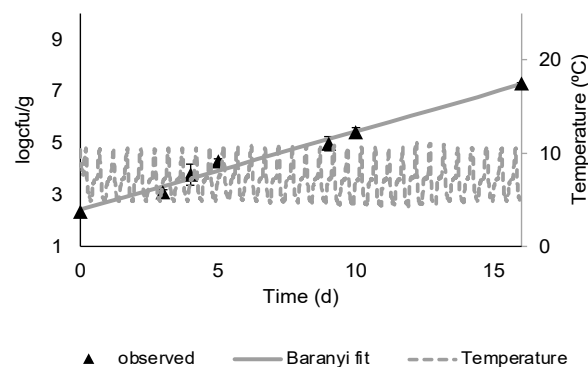


Figure 2. *M. morganii* growth in aerobically packed mullet stored at dynamic conditions ($T_{eff} = 7.4\text{ }^{\circ}\text{C}$) ($A_f = 1.0462$, $B_f = 1.0042$, $RE = -9.00$ to 10.81).

During the kinetic study of *M. morganii* growth in mullet flesh, histamine formation was also monitored. The results of histamine concentrations in aerobically packed mullet flesh stored at constant conditions of 0–20 $^{\circ}\text{C}$ are shown in Figure 3. A sigmoid equation was also fitted to the experimental data using the DMFit program. No histamine was detected at 0 and 5 $^{\circ}\text{C}$ within 1 month of storage ($p > 0.05$). This agrees with the observation for no or slow *M. morganii* growth in mullet flesh at 0 and 5 $^{\circ}\text{C}$, respectively (Figure 1). Histamine formation was observed at storage temperatures higher than 10 $^{\circ}\text{C}$, as illustrated in Figure 3. Histamine was initially detected at low concentrations (0.5–1 mg/kg) after 4, 1.5, and 1 day at 10, 15, and 20 $^{\circ}\text{C}$. Based on the microbial data presented in Figure 1, the rapid production of histamine at 10, 15, and 20 $^{\circ}\text{C}$ corresponds to microbial loads of 5–6 log cfu/g for *M. morganii*, values which were not reached after one month of storage at 0 and 5 $^{\circ}\text{C}$. In the temperature range of 10–20 $^{\circ}\text{C}$, significant histamine formation was observed ($p < 0.05$), with final histamine concentrations of 86–92 mg/kg after 16, 9, and 5 days at 10 $^{\circ}\text{C}$, 15 $^{\circ}\text{C}$, and 20 $^{\circ}\text{C}$, respectively.

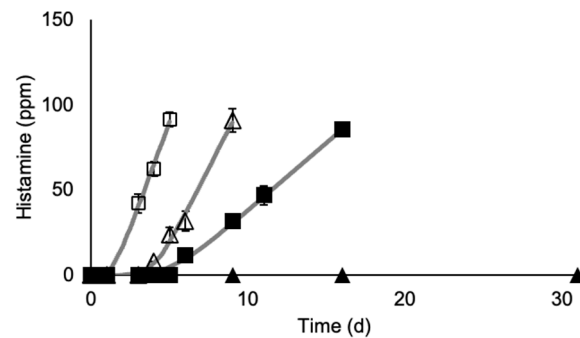


Figure 3. Histamine formation in aerobically packed mullet flesh during isothermal storage at ▲ 0 °C, ○ 5 °C, ■ 10 °C, △ 15 °C, and □ 20 °C (experimental points and predictions are based on the Baranyi Growth Model).

Kinetic data of histamine formation (Table 2) were fitted to the Arrhenius Equation (1) and the E_a values, k_{ref} , and λ_{ref} ($T_{ref} = 8$ °C) were calculated. For the growth rates k , $E_{a,k}$ was calculated as 79.3 kJ/mol and k_{ref} ($T_{ref} = 8$ °C) as 6.7 d^{−1}. For the λ value, $E_{a,\lambda}$ was calculated as 90.0 kJ/mol and λ_{ref} ($T_{ref} = 8$ °C) as 7.5 d for the temperature range of 10–20 °C.

Table 2. Kinetic parameters (production rate, k , and lag phase, λ) of histamine formation in aerobically packed mullet flesh during isothermal storage at 0–20 °C. (Average values ± Standard error).

Temperature (°C)	k (d ^{−1})	λ (d)
0	-	-
5	-	-
10	7.946 ± 0.465	4.921 ± 0.515
15	17.883 ± 3.218	3.964 ± 0.277
20	25.021 ± 2.233	1.328 ± 0.281

Figure 4 presents the comparison between observed and predicted histamine formation under dynamic conditions. The calculated RE values revealed that all points fall within the acceptable prediction zone of ±20% as defined in the literature [39,40], indicating that the developed mathematical models can be adequately used for the prediction of histamine formation in chilled mullet at variable temperature conditions. The A_f and B_f values indicated that there was a satisfactory agreement between predicted and observed histamine levels. The B_f values were within the boundaries of 0.7 (fail—safe) to 1.15 (fail—dangerous) [41].

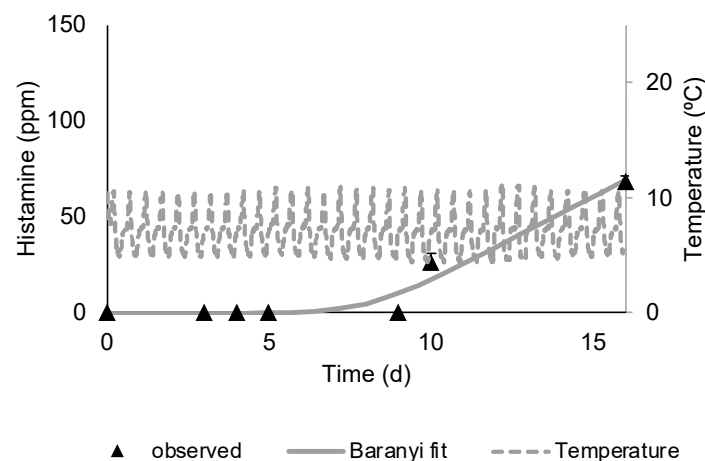


Figure 4. Histamine formation in aerobically packed mullet stored at dynamic conditions ($T_{eff} = 7.4$ °C) ($A_f = 1.1419$, $B_f = 0.8866$, $RE = -1.22$ to 18.79).

Enterobacteriaceae, in general, are often reported as the bacteria responsible for histamine production in fish. *M. morganii* has consistently been responsible for producing high levels of histamine in cultures and plays a crucial role in the formation and accumulation of histamine during the storage of fish meat [13,36]. In this study, the inoculation of samples with *M. morganii* resulted in rapid histamine formation in mullet flesh stored at refrigerated temperatures at 10, 15, and 20 °C. Histamine levels reached 50 mg/kg in mullet flesh after 12, 7, and 3 d at 10, 15, and 20 °C, respectively. The presence of *Morganella* spp. has been reported in scombroid fish, such as mullets of the genus *Mugil*, for several outbreaks of intoxication [11]. In the study reported by Trevisani et al. [11], histamine formation in mullets from the genera *Mugil* and *Liza* was attributed to *Photobacterium damsela*, which is one of the most prominent emerging pathogens for marine fish, with significant geographical distribution during the last few years [37]. In a previous study by da Costa Silva Andrade et al. [42], histamine did not significantly increase in mullet (*Mugil platanus*) stored at 0 °C [42]. Chakrabarti [43] and Kung et al. [44] reported the potential of histamine formation in commercial mullet muscle and salted mullet roes, respectively [43,44]. Tsironi et al. reported that in contaminated fish samples, *M. morganii* interacted synergistically with the natural microflora, enhancing histamine formation during storage at 10 °C [12]. In this study, *M. morganii* did not grow at melting ice temperatures (i.e., at the recommended fish storage conditions). Likewise, previous studies have shown positive correlations between microbial counts and the formation of biogenic amines, such as histamine, in fish [45]. Contamination of fish with bacteria relevant to histamine production may occur after harvesting, especially due to insufficient hygiene procedures during the handling, processing, and storage of fish. Following industry guidelines is crucial during fish processing and manufacturing to reduce microbial contamination, ensure proper handling, and maintain chilling temperatures to below 4 °C [46,47]. Implementing good hygienic practices helps prevent contamination during handling, distribution, and storage, contributing to the safety, high quality, and adequate shelf life of fish [16].

3.2. Shelf Life Evaluation of Chilled, Aerobically Packed Mullet

The development of the total viable count (TVC), *Pseudomonas* spp., and Enterobacteriaceae in aerobically packed mullet stored at constant storage temperatures of 0, 2.5, 5, 10, and 15 °C is illustrated in Figure 5a–c. The microbial growth data were adequately described by the Baranyi Growth Model ($R^2 > 0.94$ for all the tested microorganisms and storage temperatures). The initial microbial loads of the mullet flesh were 5.0 ± 0.1 , 3.7 ± 0.04 , and 3.02 ± 0.02 log cfu/g for the TVC, *Pseudomonas* spp., and Enterobacteriaceae, respectively. The initial microbial load was similar to the initial TVC populations reported for processed (gutted, sliced, or filleted) Mediterranean fish, such as gilthead seabream and European sea bass, as well as for mullet, for which the TVC was ca. 4.7 log cfu/g according to El-Ghareeb et al. [48]. The rate of microbial growth rose significantly with increases in storage temperature. ($p < 0.05$), as presented in Table 3.

Spoilage of mullet during refrigeration was attributed to both *Pseudomonas* spp. and Enterobacteriaceae growth, as the maximum microbial population (N_{max}) reached values of 8.3 ± 0.8 and 8.1 ± 1.1 log cfu/g for *Pseudomonas* spp. and Enterobacteriaceae, respectively, at all tested storage temperatures at the end of storage (0–15 °C). This was non-significantly different from the final TVC load (i.e., 9.3 ± 1.2) ($p > 0.05$). Microbial growth rates of *Pseudomonas* spp. and Enterobacteriaceae at each storage temperature were non-significantly different from the respective TVC growth rates ($p > 0.05$), indicating that *Pseudomonads* and Enterobacteria were the most important bacteria involved in the spoilage mechanism of refrigerated, aerobically packed mullet slices when samples were stored at 0–15 °C. The growth rate variables for the microorganisms tested were calculated by fitting the microbial growth rates to the Arrhenius equation (Equation (1)) and are presented in Table 4.

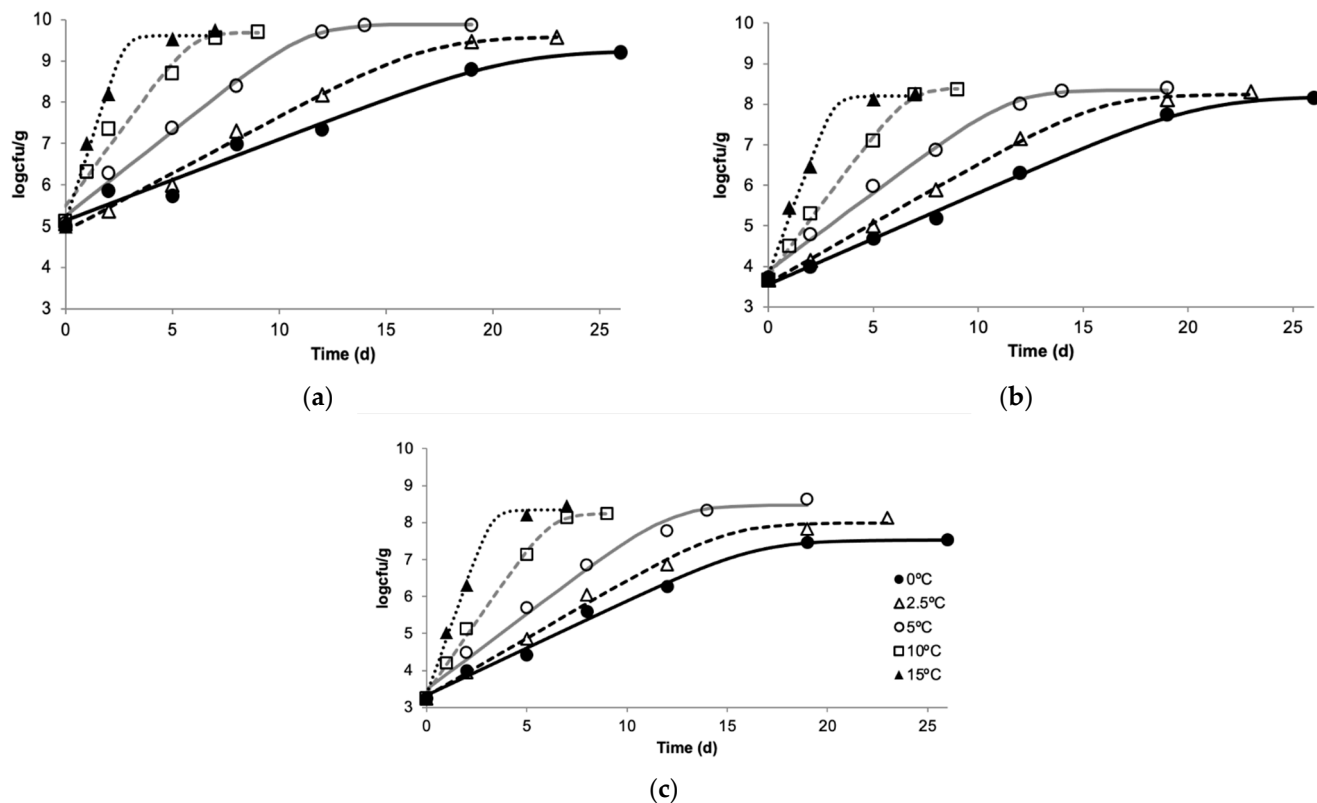


Figure 5. Development of (a) total viable count, (b) *Pseudomonas* spp., and (c) Enterobacteriaceae spp. in aerobically packed mullet slices during isothermal storage at ● 0 °C, △ 2.5 °C, ○ 5 °C, □ 10 °C, and ▲ 15 °C (experimental points and predictions based on the Baranyi Growth Model).

Table 3. Growth rate (k in d^{-1}) of TVC, *Pseudomonas* spp., and Enterobacteriaceae in aerobically packed mullet slices during isothermal storage at 0–15 °C. (Average values $\pm$ Standard error).

Temperature (°C)	k_{TVC} (d^{-1})	$k_{Pseudomonas}$ (d^{-1})	$k_{Enterobacteriaceae}$ (d^{-1})
0	0.225 ± 0.031	0.198 ± 0.044	0.215 ± 0.026
2.5	0.314 ± 0.035	0.294 ± 0.030	0.276 ± 0.028
5	0.411 ± 0.042	0.383 ± 0.052	0.409 ± 0.039
10	0.782 ± 0.056	0.693 ± 0.041	0.708 ± 0.053
15	1.624 ± 0.295	1.393 ± 0.126	1.533 ± 0.237

Table 4. Kinetic parameters E_a and k_{ref} ($T_{ref} = 4$ °C) for the growth of TVC, *Pseudomonas* spp., and Enterobacteriaceae in aerobically packed mullet slices during isothermal storage at 0–15 °C. (Average values $\pm$ Standard error).

	E_a (kJ/mol)	k_{ref} ($T_{ref} = 4$ °C) (d^{-1})
TVC	89.9 ± 6.7	0.344 ± 0.042
<i>Pseudomonas</i> spp.	79.2 ± 7.4	0.355 ± 0.051
Enterobacteriaceae	79.4 ± 8.2	0.389 ± 0.770

The sensory scores (i.e., appearance, odor, and overall impression) of the aerobically packed mullet under constant storage at 0, 2.5, 5, 10, and 15 °C are illustrated in Figure 6a–c. The decrease in sensory scores was adequately described by zero-order lines ($R^2 > 0.95$ for all tested sensory parameters and storage temperatures). The rates of decrease for the sensory scores of refrigerated, aerobically packed mullet slices are presented in Table 5. The calculated rates of decrease for the sensory scores were significantly increased ($p < 0.05$)

with increasing storage temperatures. The calculated kinetic parameters for the sensory evolution are presented in Table 6, as determined by fitting the data presented in Table 5 to the Arrhenius model (Equation (1)). The rates of decrease of the sensory scores had similar E_a values to the microbial growth rates ($p > 0.05$), indicating a strong correlation and similar temperature sensitivity. The total microbial load reached 10^8 cfu/g, which coincided with sensory rejection (overall impression score equal to 5) and the end of the shelf life. The shelf life was estimated at 15, 11, 7, 3, and 1.5 days at 0, 2.5, 5, 10, and 15 °C, respectively.

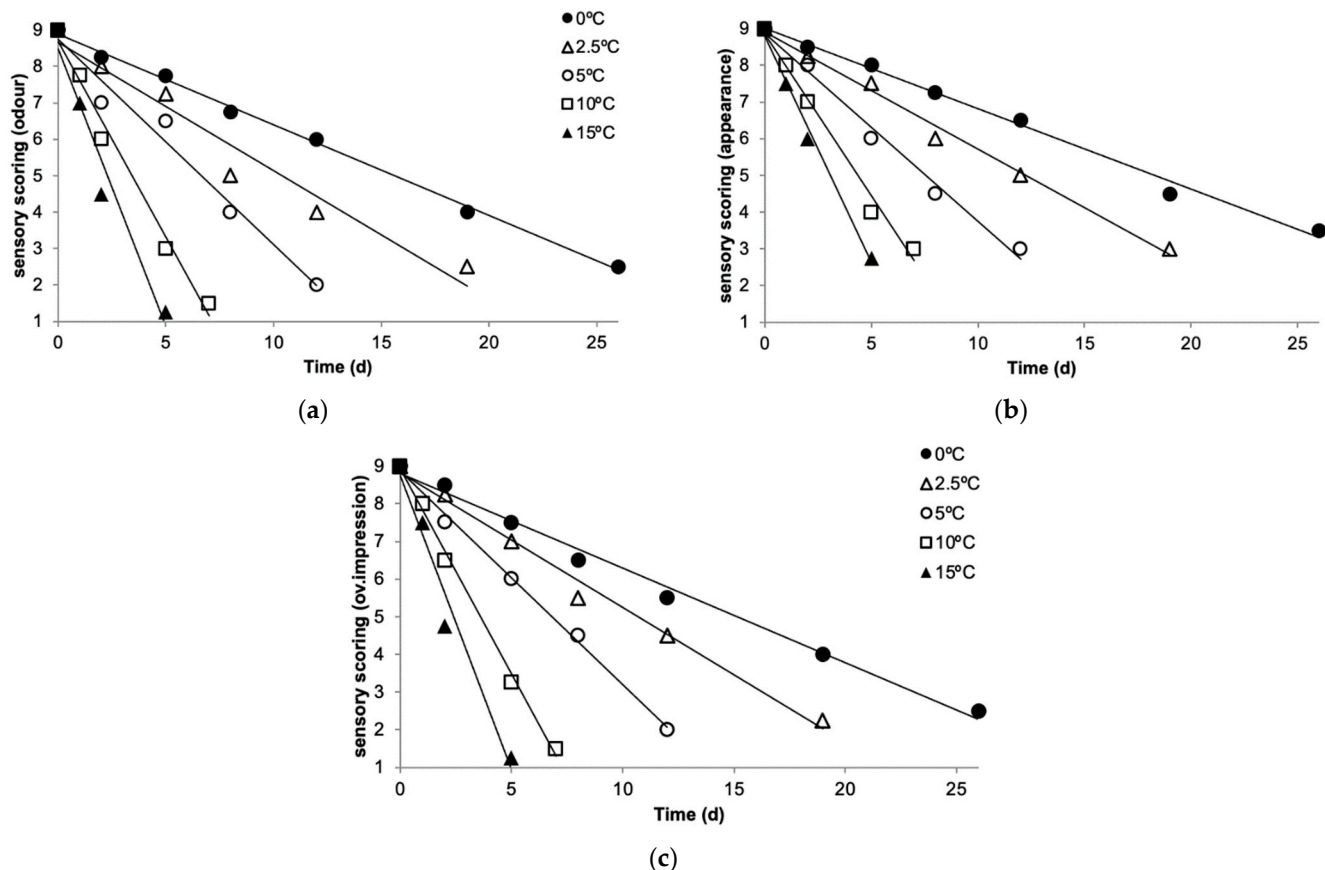


Figure 6. Sensory scoring: (a) appearance, (b) odor, and (c) overall impression of aerobically packed mullet slices during isothermal storage at ● 0 °C, △ 2.5 °C, ○ 5 °C, □ 10 °C, and ▲ 15 °C.

Table 5. Sensory scoring (k in d^{-1}): rates of decrease for odor, appearance, and overall impression, and total volatile basic nitrogen (TVBN) increase rates (k in d^{-1}) in aerobically packed mullet slices during isothermal storage at 0–15 °C. (Average values $\pm$ Standard error).

Temperature (°C)	k_{odor} (d^{-1})	$k_{appearance}$ (d^{-1})	$k_{ov.impression}$ (d^{-1})	k_{TVBN} (d^{-1})
0	0.249 ± 0.012	0.218 ± 0.017	0.251 ± 0.030	0.063 ± 0.007
2.5	0.352 ± 0.009	0.318 ± 0.026	0.357 ± 0.019	0.079 ± 0.008
5	0.564 ± 0.021	0.512 ± 0.045	0.566 ± 0.027	0.109 ± 0.009
10	1.074 ± 0.082	0.882 ± 0.113	1.088 ± 0.032	0.202 ± 0.016
15	1.518 ± 0.101	1.232 ± 0.098	1.554 ± 0.143	0.267 ± 0.022

The findings of this study align with previous data reported in the literature. da Costa Silva Andrade et al. [42] reported a limit of acceptability of 7 log-cfu/g for the mesophilic count in mullet (*Mugil platanus*) stored at 0 °C, corresponding to a shelf life of 20 days [42]. Saadia et al. [49] reported a shelf life of 14 days for mullet stored on ice (0 °C) [49]. More specifically, a high quality grade was reported for fish stored on ice for up to 4 days, while an acceptable quality grade was reported for storage longer than 4 days. Pilavtepe-Celik

et al. [50] investigated the changes in the color and chemical and sensory parameters of mullet fillets during storage at 0 °C and reported unacceptable TVB-N concentrations after 11 days and lipid oxidation after 6 days [50]. The differences in the reported values could be attributed to factors such as fish species, temperature, muscle orientation, fat content, and the presence or absence of skin in the tested samples [47].

Table 6. Kinetic parameters E_a and k_{ref} ($T_{ref} = 4$ °C) for decreases in sensory scores for the appearance, odor, and overall impression of aerobically packed mullet slices during isothermal storage at 0–15 °C. (Average values $\pm$ Standard error).

	E_a (kJ/mol)	k_{ref} ($T_{ref} = 4$ °C) (d^{-1})
Appearance	76.1 ± 5.6	0.393 ± 0.024
Odor	80.9 ± 8.4	0.447 ± 0.033
overall impression	81.4 ± 9.8	0.452 ± 0.056
TVBN	65.8 ± 8.2	0.132 ± 0.015

The development pattern of total volatile basic nitrogen (TVB-N) in the aerobically packed mullet during constant storage at 0, 2.5, 5, 10, and 15 °C is illustrated in Figure 7. TVB-N values increased over storage time and were modeled using apparent first-order equations ($R^2 > 0.98$ for all storage temperatures). The calculated kinetic parameters for TVB-N development are presented in Table 6, as determined by fitting the data presented in Table 5 to the Arrhenius model (Equation (1)). Overall impression scores were correlated with TVB-N values as follows: At temperatures of 0–5 °C, sensory rejection (with an overall impression score of 5) occurred at the same time that the TVB-N concentration reached 10.2–12.3 mg·100 g^{−1}. At higher temperatures of 10–15 °C, the samples were sensorially rejected before the development of TVB-N. A TVB-N value of 10 mg·100 g^{−1} flesh has been considered the threshold of acceptability for certain fish species [51]. Higher values for TVB-N (initial and during storage) were reported for striped, red mullet [52]. In this study, TVB-N values were very low (4.6 mg 100 g^{−1}) at the onset of the storage of the samples, indicating the increased freshness of the samples. At the end of the storage experiments at 0–15 °C, the TVB-N levels of samples were between 25–30 mg 100 g^{−1}, approaching the limit of 30 mg 100 g^{−1} imposed by European Commission Decision 95/149/EC.

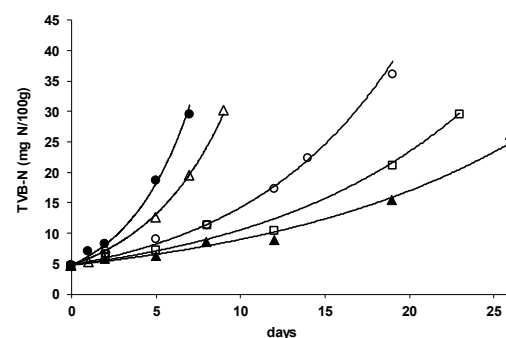


Figure 7. Development pattern of the volatile base (TVB-N) during storage of aerobically packed mullet slices during isothermal storage at $\blacktriangle$ 0 °C, $\square$ 2.5 °C, $\circ$ 5 °C, $\triangle$ 10 °C, and $\bullet$ 15 °C.

In Figure 8, the required time to reach 50 mg/kg histamine in mullet flesh and the shelf life of aerobically packed mullet slices during constant storage at 0–20 °C are illustrated. It is evident that at the recommended storage temperature range, sensory rejection will be observed well before the histamine concentration reaches 50 mg/kg fish flesh. However, when abusive temperatures prevail, histamine should be considered a risk for human consumption of mullet as the required time to reach 50 mg/kg histamine approaches the time to the end of the shelf life of the fish.

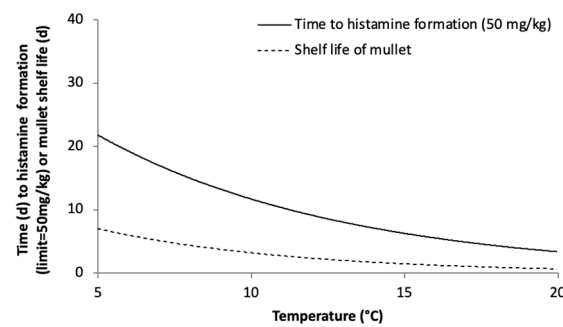


Figure 8. Time required to reach a histamine concentration of 50 mg/kg in mullet flesh and the shelf life of aerobically packed mullet slices during isothermal storage at 0–20 °C.

In order to illustrate the applicability of the developed validated predictions, a theoretical time–temperature framework for the transportation and storage of fish has been considered, as illustrated in Figure 9a,b. This scenario includes temperature fluctuations that may occur in the actual supply chain of fish, considering all the stages from harvesting, transportation, and storage at the processing plant, followed by processing and packaging, transportation to the distribution center, and retail display, as well as subsequent transportation and storage at the consumer level. The temperature range for this scenario ranges from 0.1 to 17.4 °C. Assuming a lower initial microbial load (i.e., 2.5 log cfu/g for TVC), which has also been reported in the literature for freshly caught fish [53], and an initial histamine concentration in fish flesh of 0 ppm, the predictive models developed within the present study result in the development of TVC and histamine as presented in Figure 9a,b, respectively. During fish storage under the selected time–temperature scenario, significant temperature fluctuations occur that result in quality deterioration due to the growth of spoilage bacteria (TVC after 48 h is calculated to be 3.7 log cfu/g, corresponding to a remaining shelf life of 12 days at 4 °C), while the end of shelf life (corresponding to a TVC value limit of 8 log cfu/g) is calculated as 240 h. At this stage, the histamine level is calculated as 50 mg, indicating a public health risk for histamine due to mullet consumption under the tested time–temperature scenario, as the level is getting closer to the end of the shelf life.

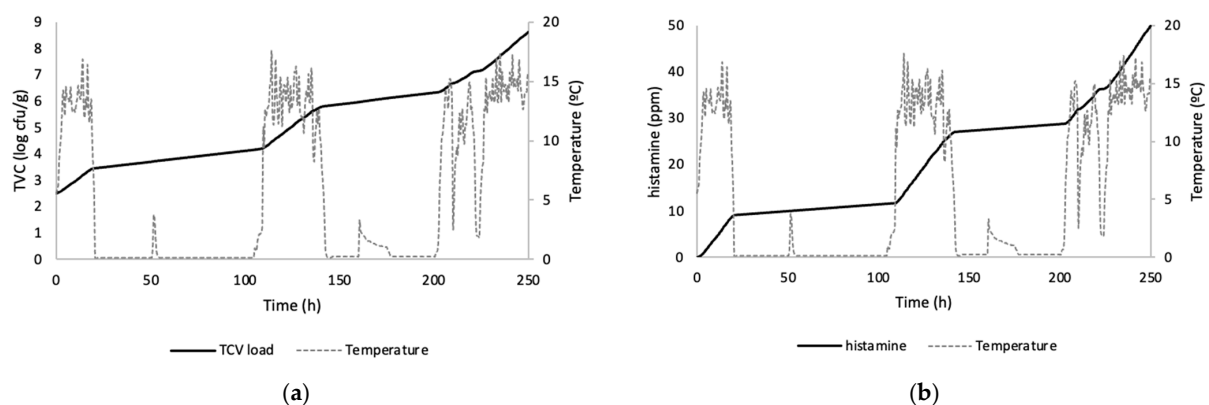


Figure 9. (a) Total viable count (TVC) growth (log cfu/g) and (b) histamine concentration (ppm) in aerobically stored mullet under a theoretical time–temperature scenario for transportation and storage with temperature fluctuations.

4. Conclusions

The objective of this work was to evaluate and quantitatively describe the deterioration in the quality of chilled mullet based on histamine formation (as a result of *M. organii* growth) at constant and dynamic storage conditions (temperature range: 0–20 °C). The quality/safety parameters were defined and the threshold values corresponding to sensory

rejection (i.e., end of shelf life) or safety limits (i.e., histamine concentration in fish flesh) were determined. Predictive models for histamine formation and shelf life estimation were developed. This study revealed that at temperatures below 5 °C, sensory rejection occurs before histamine levels reach the critical concentration of 50 mg/kg in fish flesh, marking the end of shelf life. However, at abusive temperatures (>5 °C and especially close to 20 °C), histamine production becomes a significant risk for human consumption due to the rapid growth of *M. morganii*. Importantly, no histamine formation was observed at temperatures below 10 °C, emphasizing that maintaining low storage temperatures is crucial for controlling histamine production and ensuring the safety and quality of mullet. One aspect of this study is that it focused on *M. morganii* growth in sterile fish flesh, simulating a worst-case scenario. While this offers valuable insights into histamine formation, it does not consider the influence of other microbial flora commonly present in fish. Future research could examine how mixed microbial communities affect histamine production and spoilage at both optimal and abusive temperatures. Additionally, exploring the impact of preservation methods, such as modified atmosphere packaging (MAP) or natural preservatives, could provide further understanding of how to control histamine formation and extend the shelf life of mullet under various storage conditions. Uncontrolled variables that could influence this study include the initial microbial load, oxygen exposure, and fish stress. Variations in handling, hygiene, and environmental conditions before storage can affect microbial growth and spoilage. Oxygen availability, influenced by packaging and handling, may also impact histamine production, especially with temperature fluctuations. These findings highlight the practical importance of stringent storage regulations, as low-temperature control is essential to ensure food safety and protect consumers from histamine-related health risks in fish products. With the predictive models developed in this study, the seafood industry can better monitor and manage storage conditions to prevent histamine-related foodborne illness. Ultimately, this work emphasizes the urgent need for robust food safety protocols and storage regulations. Growth models of *M. morganii* allow prediction of, e.g., the effect of delayed chilling, where products can sometimes be exposed to storage at ambient temperatures followed by chilled storage at 0–5 °C.

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